Practical-1

Solution of Cauchy problem for First-Order PDE

Problem: To obtain the solution of a Cauchy problem for the given first-order PDE

$$u_x - u_y = 1$$

with the Cauchy data $u(x, 0) = x^2$.

Solution:

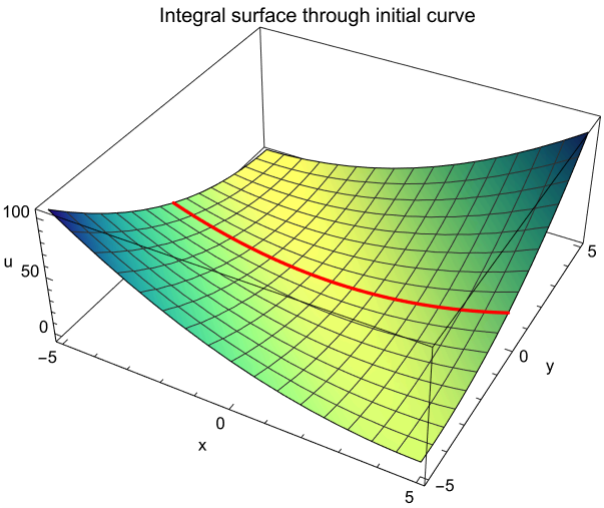
In …

```
pde = D[u[x, y], x] - D[u[x, y], y] == 1

initialCondition = u[x, 0] == x^2;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    Evaluate[u[x, y] /. solution], {x, -5, 5}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve"
];

initialCurve = ParametricPlot3D[
    {t, 0, t^2}, {t, -5, 5},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$-u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$

$\{\{u[x, y] \to x^2 - y + 2xy + y^2\}\}$



Result: The solution of the given Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Practical-2

Plotting the Characteristics for the First-Order PDE

Problem: To find the characteristics of the equation

$$(u-y)\ u_x + y\,u_y = x+y$$

and plot them.

Solution:

The auxiliary equations of the given PDE is

$$\frac{dx}{u-y} = \frac{dy}{y} = \frac{du}{x+y} \tag{1}$$

We have,

$$\Rightarrow \frac{dx+du}{(u-y)+(x+y)} = \frac{dy}{y}$$

$$\Rightarrow \frac{d(x+u)}{x+u} = \frac{dy}{y}$$

$$\Rightarrow \int \frac{d(x+u)}{x+u} = \int \frac{dy}{y}$$

$$\Rightarrow \log(x+u) = \log(y) + \log(c_1)$$

$$\Rightarrow \frac{x+u}{y} = c_1$$

$$\therefore\ u = y\,c_1 - x \tag{2}$$

Here, c_1 is a constant.
Again, we have from Eq.(1),

$$\Rightarrow \frac{dx+dy}{(u-y)+y} = \frac{du}{x+y}$$

$$\Rightarrow \frac{d(x+y)}{u} = \frac{du}{x+y}$$

$$\Rightarrow (x+y)\,d(x+y) = u\,du$$

$$\Rightarrow \int (x+y)\,d(x+y) = \int u\,du$$

$$\Rightarrow \frac{(x+y)^2}{2} = \frac{u^2}{2} + \frac{c_2}{2}$$

$$\Rightarrow (x+y)^2 - u^2 = c_2$$

$$\therefore\ u = \sqrt{(x+y)^2 - c_2} \tag{3}$$

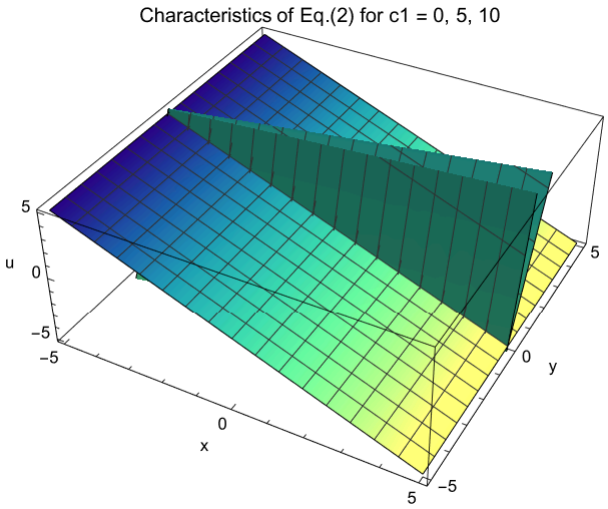
Here, c_2 is a constant.

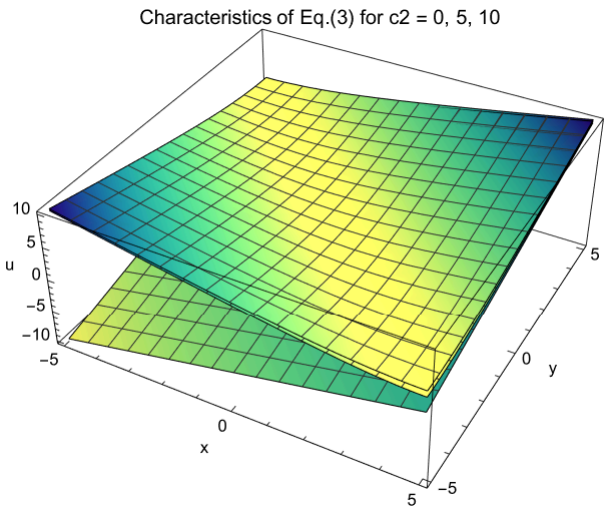
In …

```
(* for c1 = 0, 5, 10 *)
f0 = Plot3D[-x, {x, -5, 5}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(2) for c1 = 0, 5, 10"
```

```
];
f1 = Plot3D[5*y - x, {x, -5, 5}, {y, -5, 5}];
f2 = Plot3D[10*y - x, {x, -5, 5}, {y, -5, 5}];

Show[f0, f1, f2]

(* for c2= 0, 5, 10 *)
h0 = Plot3D[x + y, {x, -5, 5}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(3) for c2 = 0, 5, 10"
];
h1 = Plot3D[Sqrt[(x + y)^2 + 5], {x, -5, 5}, {y, -5, 5}];
h2 = Plot3D[Sqrt[(x + y)^2 + 10], {x, -5, 5}, {y, -5, 5}];

Show[h0, h1, h2]
```

Out…





Result: The characteristics of the given PDE are plotted in the output.

Practical-3

Plot the Integral Surfaces of a given First-Order PDE with Initial Data

Problem: To find the solution of the given PDE

$$u_x + y\,u_y = 0$$

with the Cauchy data $u(0, y) = 4y$.

Plot the integral surface with in the range $(x, -5, 5)$ & $(y, -5, 5)$.

Solution:

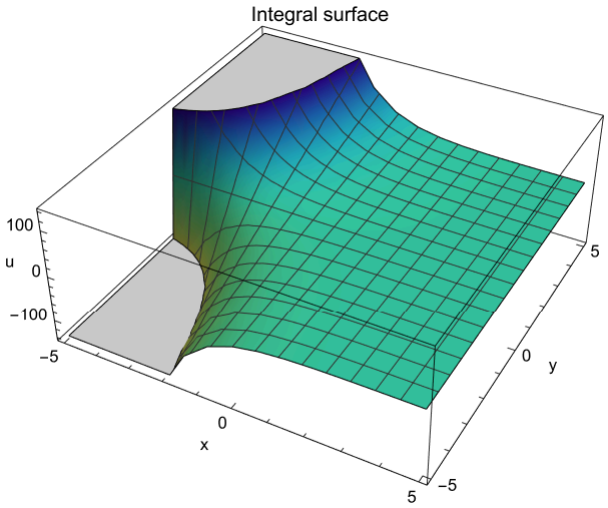
In …

```
pde = D[u[x, y], x] + y*D[u[x, y], y] == 0

initialCondition = u[0, y] == 4*y;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
     Evaluate[u[x, y] /. solution], {x, -5, 5}, {y, -5, 5},
     AxesLabel -> {"x", "y", "u"},
     PlotLabel -> "Integral surface"
]
```

Out…

$y\, u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 0$

$\{\{u[x, y] \to 4\, e^{-x}\, y\}\}$



Result: The solution of the given Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.